

The Program in Action

When you run *primenumbers.py*, the output will look like this:

```
Prime Numbers, by Al Sweigart al@inventwithpython.com
--snip--
Enter a number to start searching for primes from:
(Try 0 or 1000000000000 (12 zeros) or another number.)
> 0
Press Ctrl-C at any time to quit. Press Enter to begin...
2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71,
73, 79, 83, 89, 97, 101, 103, 107, 109, 113, 127, 131, 137, 139, 149, 151,
157, 163, 167, 173, 179, 181, 191, 193, 197, 199, 211, 223, 227, 229, 233,
239, 241, 251, 257, 263, 269, 271, 277, 281, 283, 293, 307, 311, 313, 317,
331, 337, 347, 349, 353, 359, 367, 373, 379, 383, 389, 397, 401, 409, 419,
421, 431, 433, 439, 443, 449, 457, 461, 463, 467, 479, 487, 491, 499, 503,
509, 521, 523, 541, 547, 557, 563, 569, 571, 577, 587, 593, 599, 601, 607,
613, 617, 619, 631, 641, 643, 647, --snip--
```

How It Works

The `isPrime()` function accepts an integer and returns `True` if it is a prime number. Otherwise, it returns `False`. Project 24 is worth studying if you're trying to understand this program. The `isPrime()` function essentially looks for any factors in the given number and returns `False` if it finds any.

The algorithm in this program can quickly find large prime numbers. The number one trillion has a mere 10 digits. But to find prime numbers that are as big as a googol (a one followed by 100 zeros), you need to use an advanced algorithm such as the Rabin-Miller primality test. Chapter 22 of my book *Cracking Codes with Python* (No Starch Press, 2018) has a Python implementation of this algorithm.

```
1. """Prime Numbers, by Al Sweigart al@inventwithpython.com
2. Calculates prime numbers, which are numbers that are only evenly
3. divisible by one and themselves. They are used in a variety of practical
4. applications.
5. More info at: https://en.wikipedia.org/wiki/Prime_number
6. View this code at https://nostarch.com/big-book-small-python-projects
7. Tags: tiny, math, scrolling"""
8.
9. import math, sys
10.
11. def main():
12.     print('Prime Numbers, by Al Sweigart al@inventwithpython.com')
13.     print('Prime numbers are numbers that are only evenly divisible by')
14.     print('one and themselves. They are used in a variety of practical')
15.     print('applications, but cannot be predicted. They must be')
16.     print('calculated one at a time.')
17.     print()
18.     while True:
19.         print('Enter a number to start searching for primes from:')
20.         print('(Try 0 or 1000000000000 (12 zeros) or another number.)')
```